INDUSTRIAL CARBON CREDITS CALCULATOR

Step-by-Step User Guide

For: Industrial facility managers, compliance officers, and environment teams

Standard Used: UNFCCC CDM Methodology | Data Sources: IPCC AR6, IEA, India CEA

Formula: Net CERs = Baseline Emissions − Project Emissions − Leakage

The link of Carbon Credits Calculator click on this link Carbon Credits Calculator or you can directly paste this link in your browser https://eco-carbon-path.lovable.app/

Introduction

This guide explains, step by step, how to use the Industrial Carbon Credits Calculator website. You do not need any technical background. By the end, you will know how to enter your factory's data, understand your emissions, and download an official report.

What Does This Calculator Do?

It calculates how many carbon credits your factory has earned (or needs to buy) based on your emissions data. Carbon credits are a way of measuring how much CO_2 your factory puts into the air compared to what it 'should' be putting in.

The Formula Explained Simply

Net Carbon Credits = Baseline Emissions − Project Emissions − Leakage

- Baseline Emissions — How much CO_2 your factory WOULD have emitted without any green changes
- Project Emissions — How much CO_2 your factory ACTUALLY emitted
- Leakage — CO_2 that shifts to somewhere else because of your project

If Baseline > Project, you EARNED credits. If Project > Baseline, you need to BUY credits.

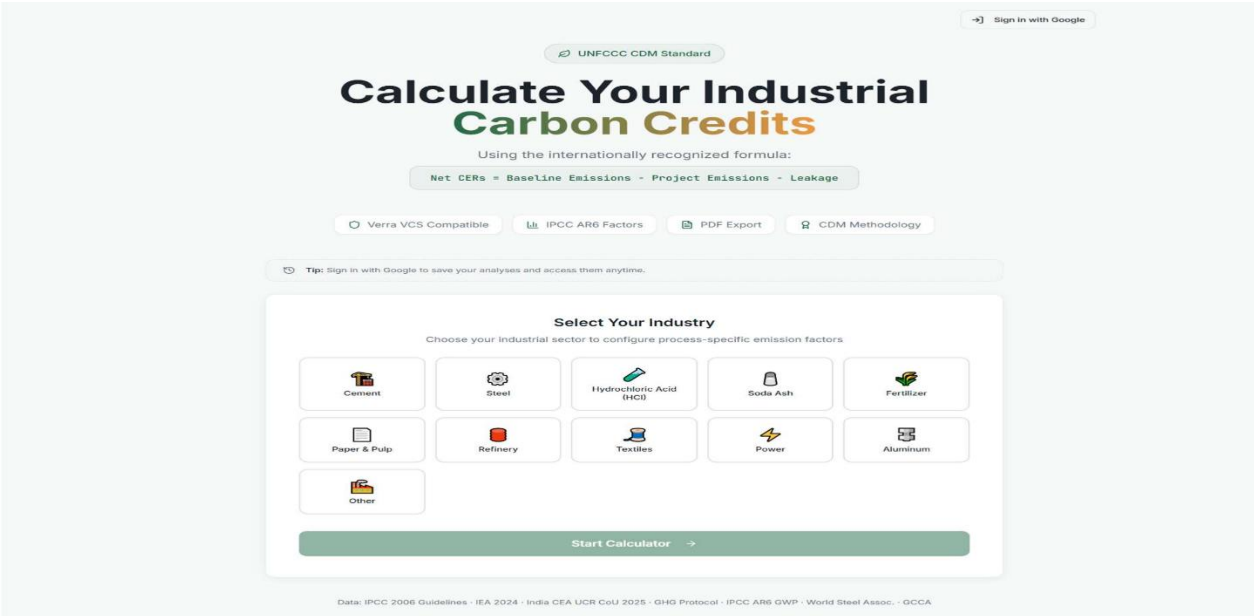
What You Will Need Before You Start

Collect these documents/numbers before you begin. This will make the process much faster:

What You Need	Where to Find It
Company name and plant/facility name	Company registration documents
Reporting year (e.g. FY2025)	Your accounting team
Months of operation (usually 12)	Plant manager
Annual production volume (tonnes)	Production records
Fuel types and quantities used (tonnes)	Fuel purchase invoices
Electricity consumed (kWh)	Electricity bills
Waste generated and type (tonnes)	Waste management records
Transport distances and cargo weights	Logistics department

Opening the Website

Open any web browser (Chrome, Edge, Firefox) on your computer. The website loads automatically and shows the home screen.



Tip — Save Your Progress

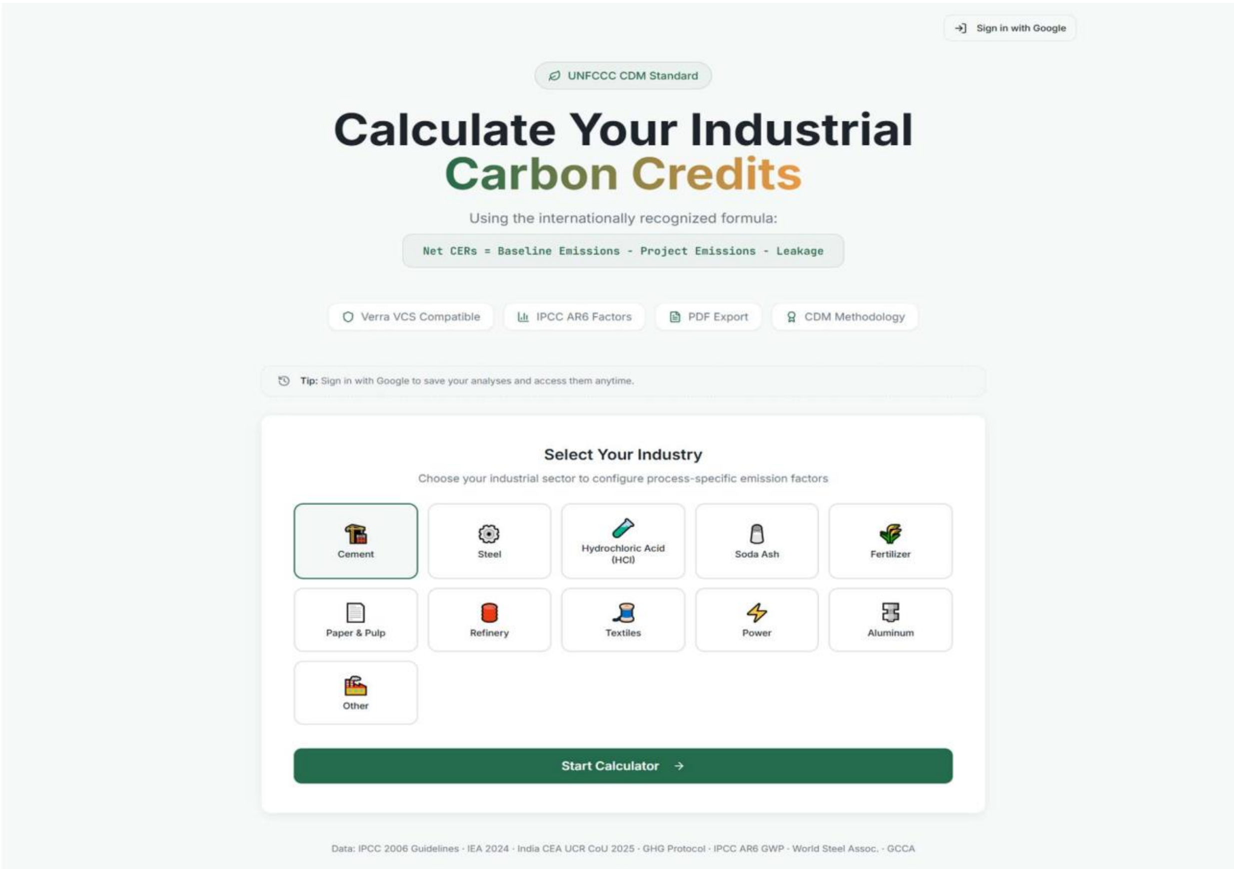
Click 'Sign in with Google' at the top-right corner of the page.
Signing in means your work is saved automatically. Without signing in,
if you close the browser, all your entered data will be lost.

What You See on the Home Screen

The home screen has two main areas:

Section	What It Means
UNFCCC CDM Standard badge (top)	Confirms this tool uses the internationally approved carbon calculation method
'Calculate Your Industrial Carbon Credits' heading	The main title — you are in the right place
Formula bar showing 'Net CERs = Baseline − Project − Leakage'	A reminder of how carbon credits are calculated
Four green badges: Verra VCS, IPCC AR6, PDF Export, CDM	Marks showing which international standards are supported

Industry selection grid (centre)	Where you choose your type of factory — THIS IS YOUR FIRST STEP
'Start Calculator' green button	Click after choosing your industry to begin



Step 1

Select Your Industry

Choose the type of factory or facility you are calculating for

On the home screen, you will see a grid of industry options. Click on the one that matches your factory:

Industry Option	Choose This If Your Factory Makes...
Cement	Cement, clinker, or related construction materials
Steel	Steel, iron, or metal products
Hydrochloric Acid (HCl)	Hydrochloric or other industrial acids
Soda Ash	Soda ash, sodium carbonate, or cleaning chemicals
Fertilizer	Fertilizers, urea, or agricultural chemicals
Paper & Pulp	Paper, cardboard, or wood pulp products
Refinery	Petroleum products, petrol, diesel, or chemicals
Textiles	Cloth, yarn, fabric, or garments
Power	Electricity generation (coal, gas, or other fuel)
Aluminum	Aluminum or aluminum alloy products
Other	Any other type of industrial facility

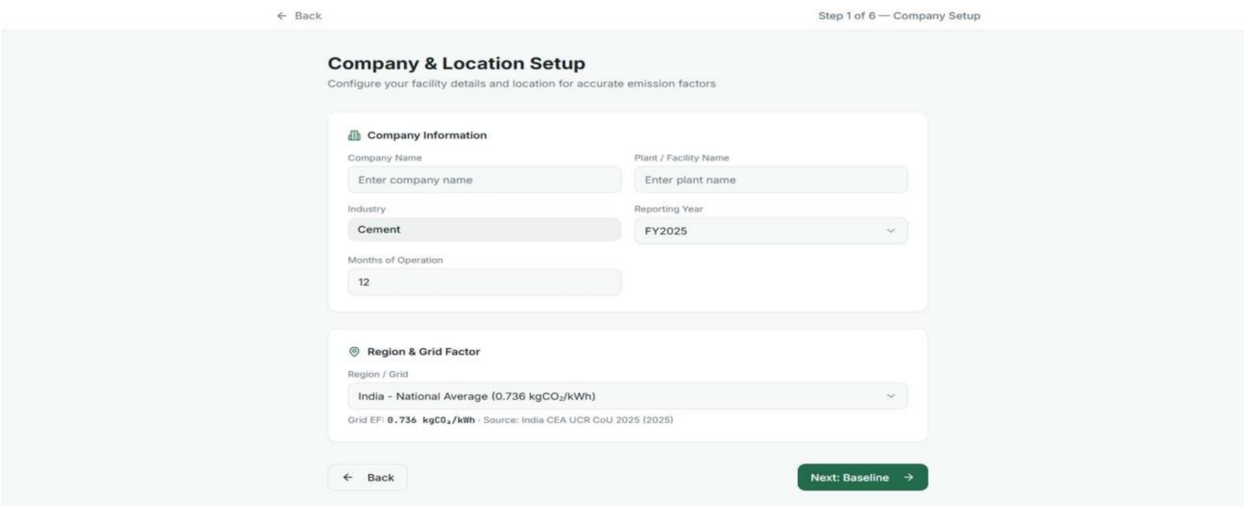
Example Used in This Guide

Throughout this guide, we use a CEMENT factory as the example shown on previous page.
Company: Ultratech Cement | Plant: ADITYA CEMENT WORKS, Chittorgarh
The steps are the same for all industries only the emission factors change.

After clicking your industry (it will show a green border around it), click the green 'Start Calculator →' button.

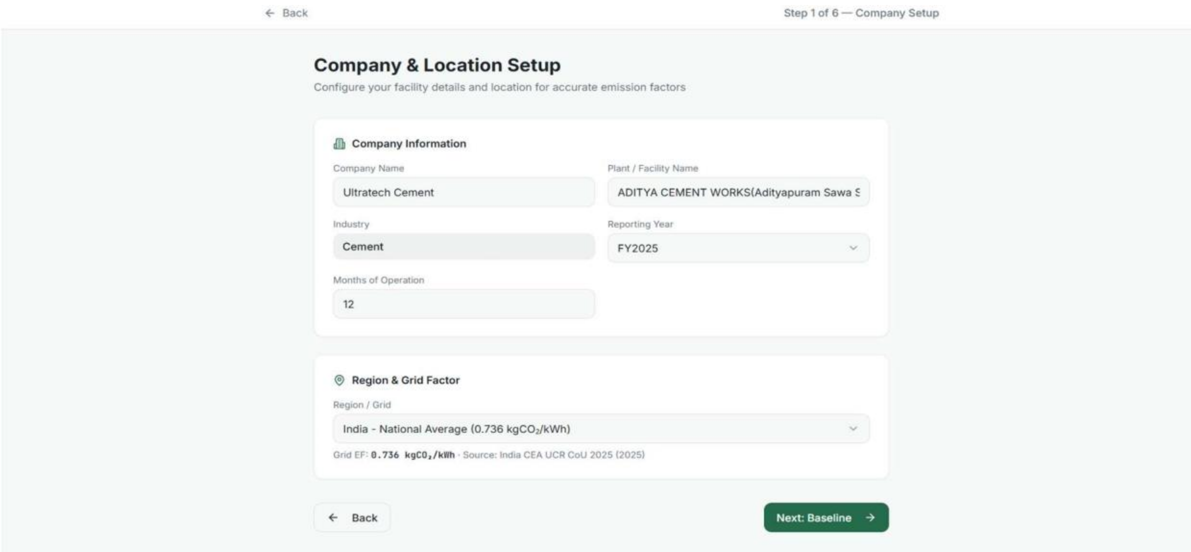
Step 2	**Company & Location Setup** Enter your factory's basic details — this is Step 1 of 6

You will see a form titled 'Company & Location Setup'. Fill in each field as described below:



Company Information Section

Field Name	What to Enter
Company Name	Your company's official name (e.g., Ultratech Cement)
Plant / Facility Name	The specific factory or plant name (e.g., ADITYA CEMENT WORKS, Chittorgarh)
Industry	This is pre-filled based on your choice in Step 1 — no need to change
Reporting Year	Select the financial year you are reporting for (e.g., FY2025)
Months of Operation	How many months the plant ran that year — usually 12



Region & Grid Factor Section

This section sets the electricity emission factor for your region. It is used to calculate how much CO_2 was produced by the electricity your factory used.

Field Name	What to Enter / Note
Region / Grid	Select your region from the dropdown. For most Indian factories, choose 'India – National Average (0.736 $kgCO_2$ /kWh)'
Grid EF (shown below the dropdown)	This fills automatically — you do not need to type anything. It shows the emission factor in kg of CO_2 per kWh of electricity.

ℹ What is a Grid Emission Factor?

When your factory uses electricity from the power grid, some CO_2 was released at the power station to generate it. The Grid EF number tells the calculator how much.
India National Average: 0.736 $kgCO_2$ /kWh (Source: India CEA UCR 2025)

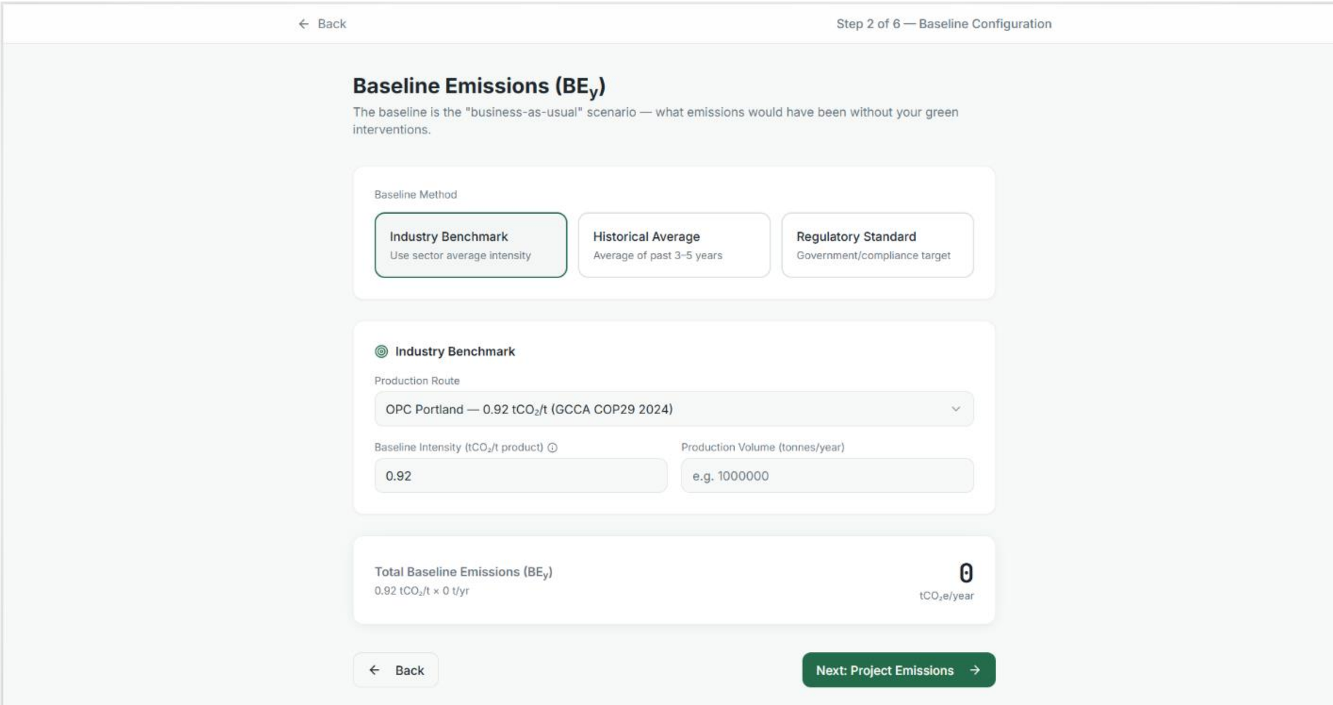
When all fields are filled, click 'Next: Baseline → to go to Step 3.

Step 3

Baseline Emissions

Set your factory's 'business-as-usual' reference — this is Step 2 of 6

The Baseline is the amount of CO_2 your factory WOULD have emitted without any energy-saving or green projects. Think of it as the 'before' number.



Choose a Baseline Method

You must select one of three methods. Click the method that fits your situation:

Baseline Method	When to Use It
Industry Benchmark (Recommended)	Use the average emission intensity for your sector. Best for most factories. The number is pre-filled based on international standards.
Historical Average	Use your own factory's past 3–5 years of emissions data. Choose this if your factory's performance is well-documented.
Regulatory Standard	Use the government-set compliance target. Choose this if you are specifically reporting for a government scheme.

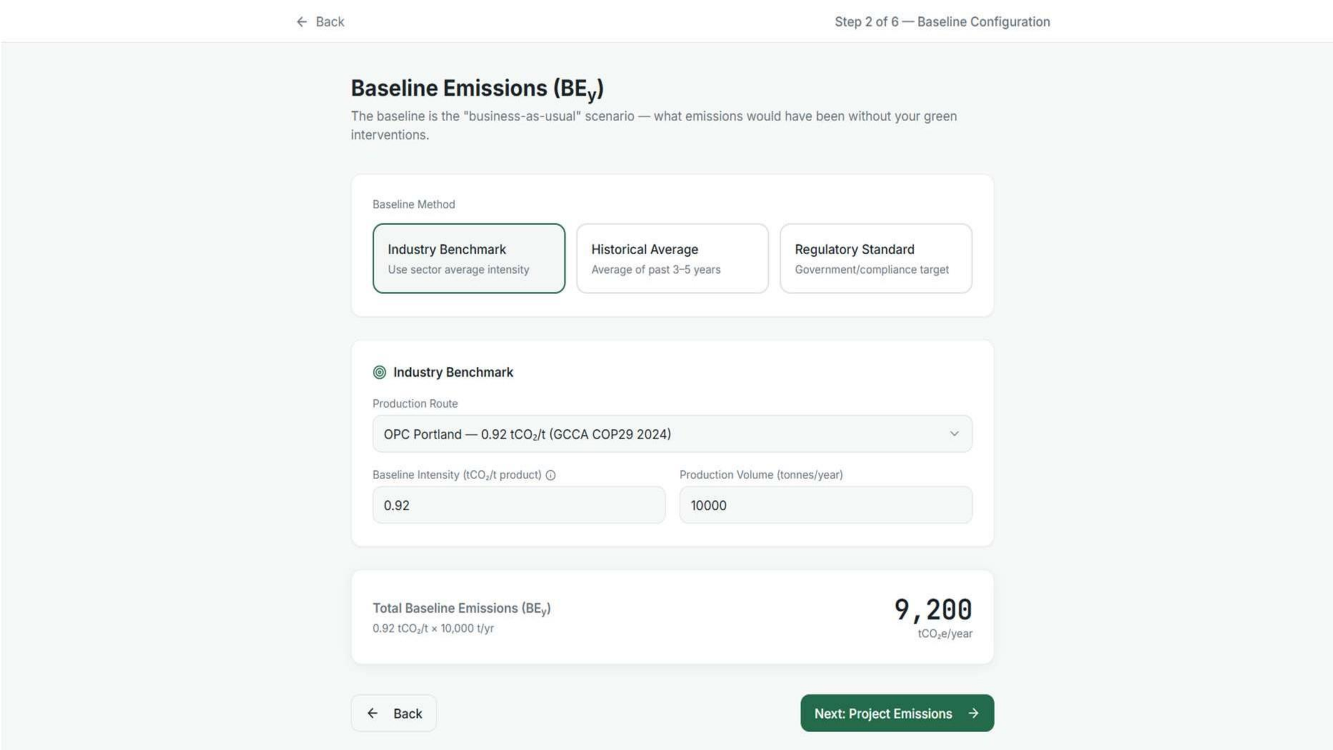
Filling In the Industry Benchmark (most common choice)

Field Name	What to Enter
Production Route (dropdown)	Select the specific production process used in your factory. Example for cement: 'OPC Portland — 0.92 tCO_2 /t (GCCA COP29 2024)'

Baseline Intensity (tCO_2 /t product)	Pre-filled automatically based on your production route. You can edit if you have certified data.
Production Volume (tonnes/year)	Enter your factory's annual production in tonnes (e.g., 10000)

Total Baseline Emissions

The calculator automatically shows 'Total Baseline Emissions (BEy)'.
Example: 0.92 × 10,000 = 9,200 tCO_2 /year
This is the reference number that your actual emissions will be compared against.



Click 'Next: Project Emissions → to move to Step 4.

Step 4

Project Emissions

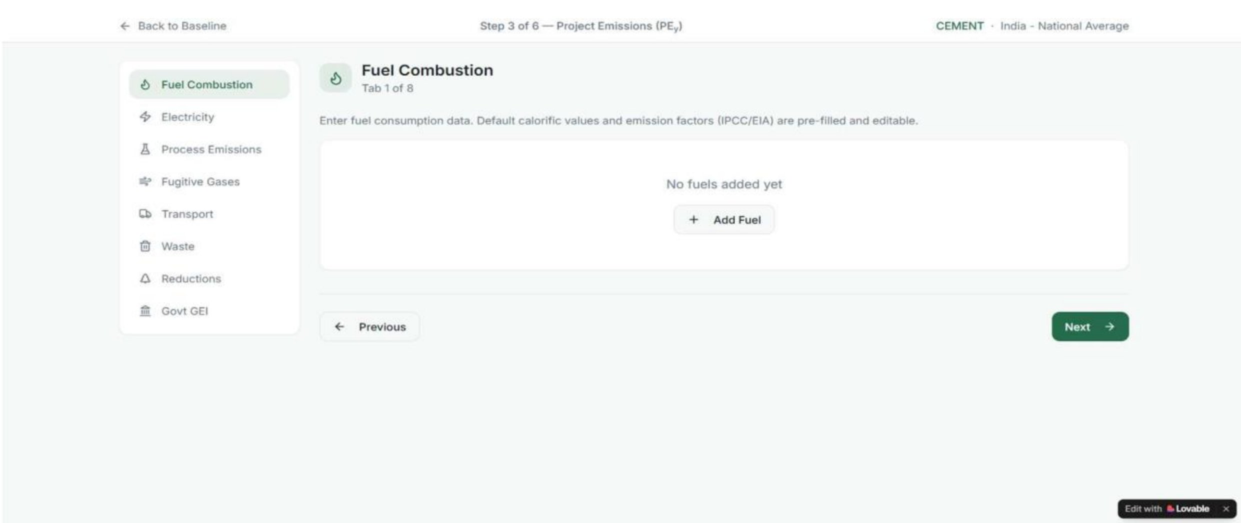
Enter all actual emissions from your factory — this is Step 3 of 6

This is the largest step. You will enter your factory's ACTUAL emissions from eight different categories. Each category is a separate tab on the left-side menu.

The 8 Emissions Tabs

Tab 1: Fuel Combustion	Tab 2: Electricity	Tab 3: Process Emissions
Tab 4: Fugitive Gases	Tab 5: Transport	Tab 6: Waste
Tab 7: Reductions	Tab 8: Govt GEI	

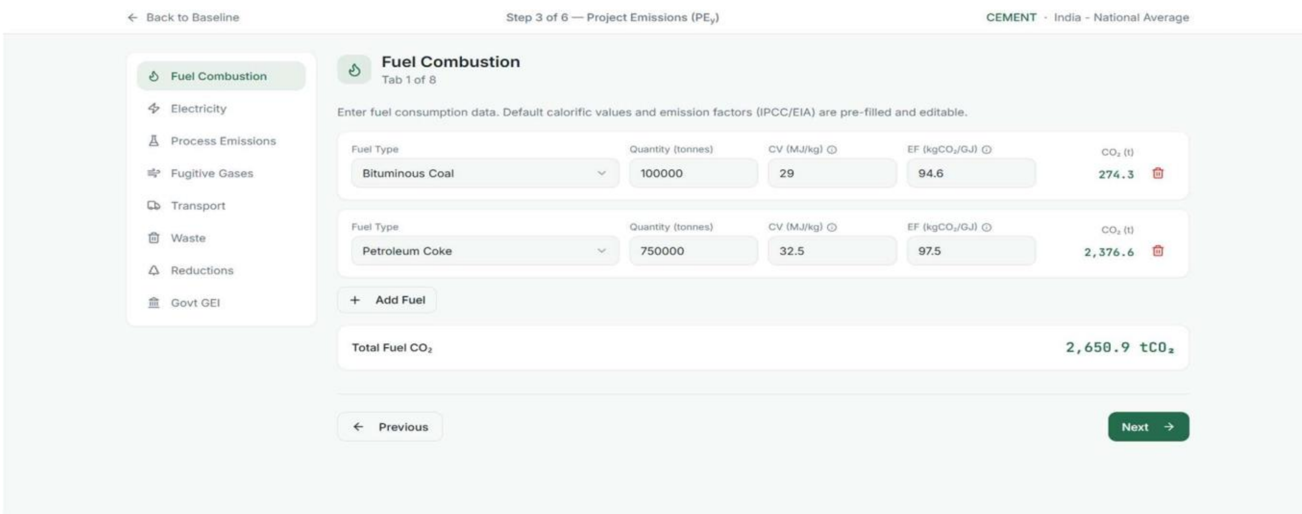
Work through each tab from top to bottom. A checkmark (✓) appears next to a tab once it is complete.



Tab 1 — Fuel Combustion (Scope 1)

Enter all fuels your factory burns directly — coal, gas, diesel, petroleum coke, etc.
Click '+ Add Fuel' to add each fuel type. For each fuel, fill in:

Field	What to Enter
Fuel Type (dropdown)	Select the fuel: Bituminous Coal, Petroleum Coke, Natural Gas, Diesel, etc.
Quantity (tonnes)	How many tonnes of that fuel were burned in the year
CV — Calorific Value (MJ/kg)	Pre-filled from IPCC/EIA data. Only change if you have a certified lab test value.
EF — Emission Factor ($kgCO_2$ /GJ)	Pre-filled from IPCC/EIA data. Only change if certified.
CO_2 (t)	Calculated automatically — you do not enter this.

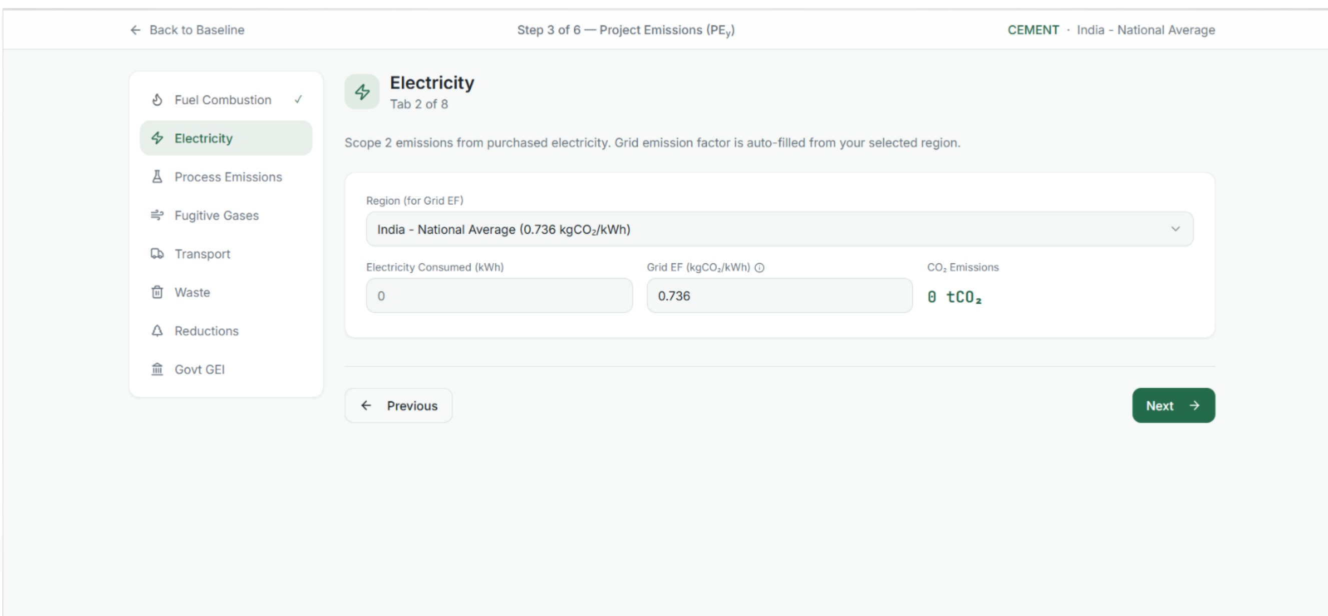


Example: Bituminous Coal — 100,000 tonnes → CO_2 = 274.3 t | Petroleum Coke — 750,000 t → CO_2 = 2,376.6 t

Total Fuel CO_2 will show at the bottom as shown above. Click 'Next →' to go to Tab 2.

Tab 2 — Electricity (Scope 2)

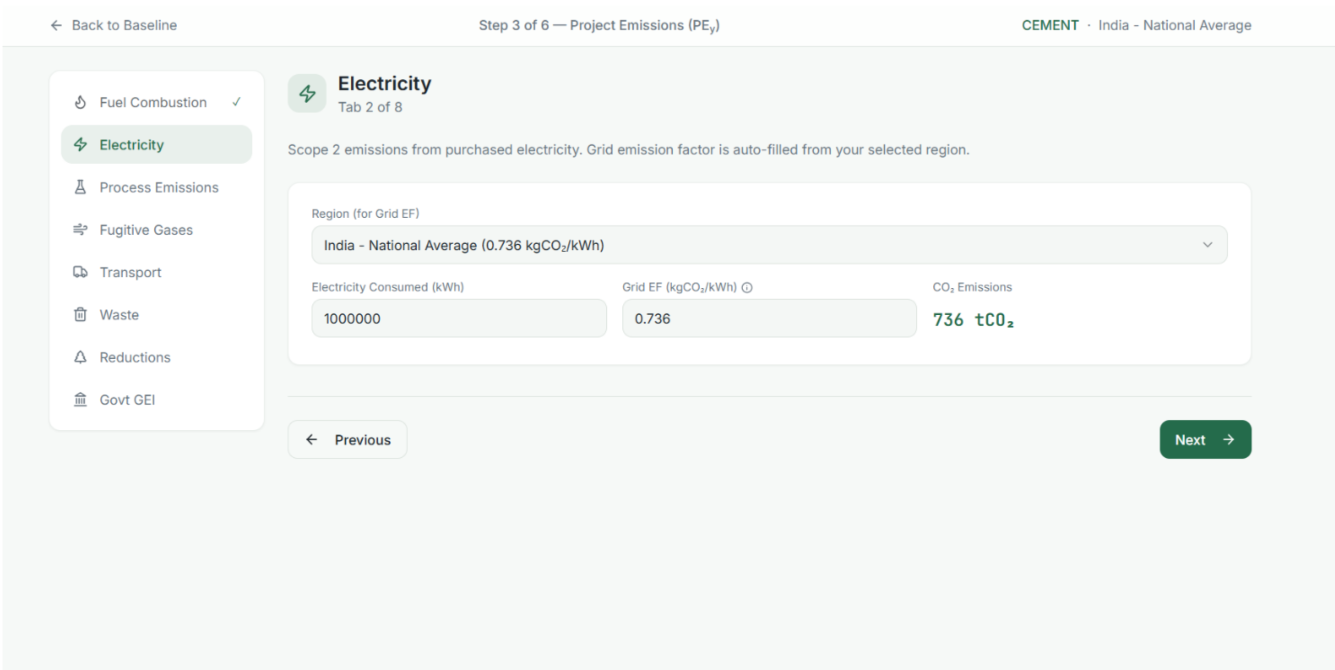
Enter the electricity your factory purchased from the power grid.



Field	What to Enter
Region (for Grid EF)	Pre-filled from Step 2. No change needed but you have to change if you want.
Electricity Consumed (kWh)	Total electricity units consumed in the year (from your electricity bills)
Grid EF ($kgCO_2$ /kWh)	Pre-filled automatically — do not change unless you have certified data
CO_2 Emissions	Calculated automatically

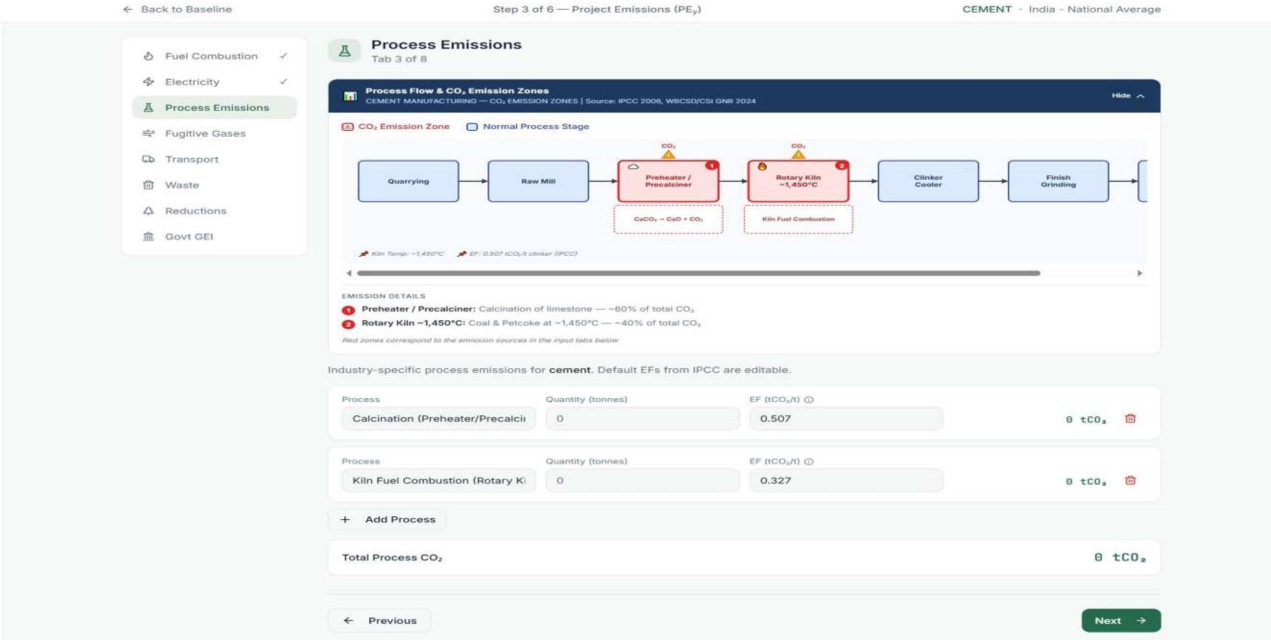
Note on Electricity

If your factory generates its own electricity (solar, captive power plant), enter only the electricity PURCHASED from the external grid. Self-generated renewable electricity does not go here it goes in the Reductions tab.



Tab 3 — Process Emissions (Scope 1)

These are emissions from chemical reactions inside your factory not from burning fuel.



Cement-Specific: The Process Flow Diagram

For cement factories, this tab shows a process flow diagram of CO_2 emission zones:

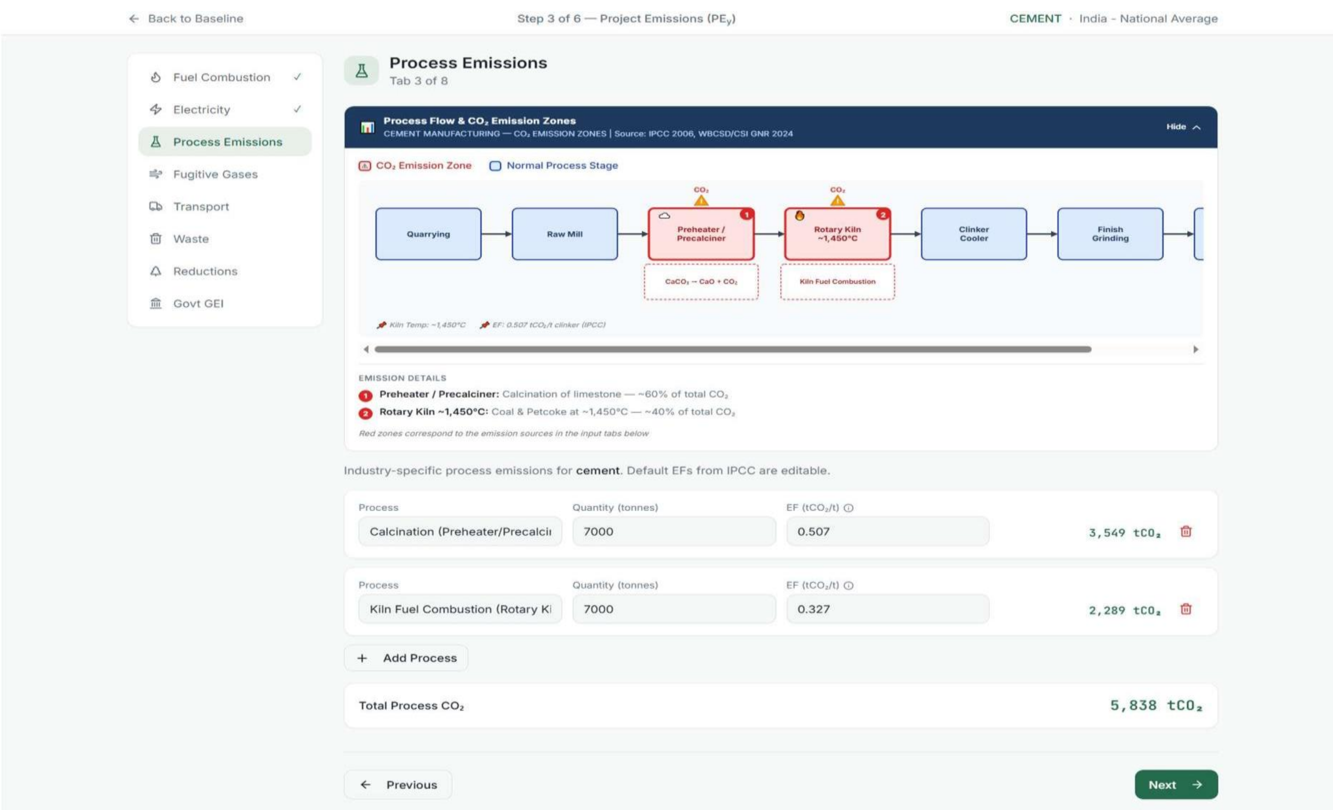
• Preheater/Precalciner — Calcination of limestone (accounts for ~60% of total CO_2)

• Rotary Kiln at 1,450°C — Coal & petcoke burning (accounts for ~40% of total CO_2)

The diagram is for reference only. Enter your data in the fields below it.

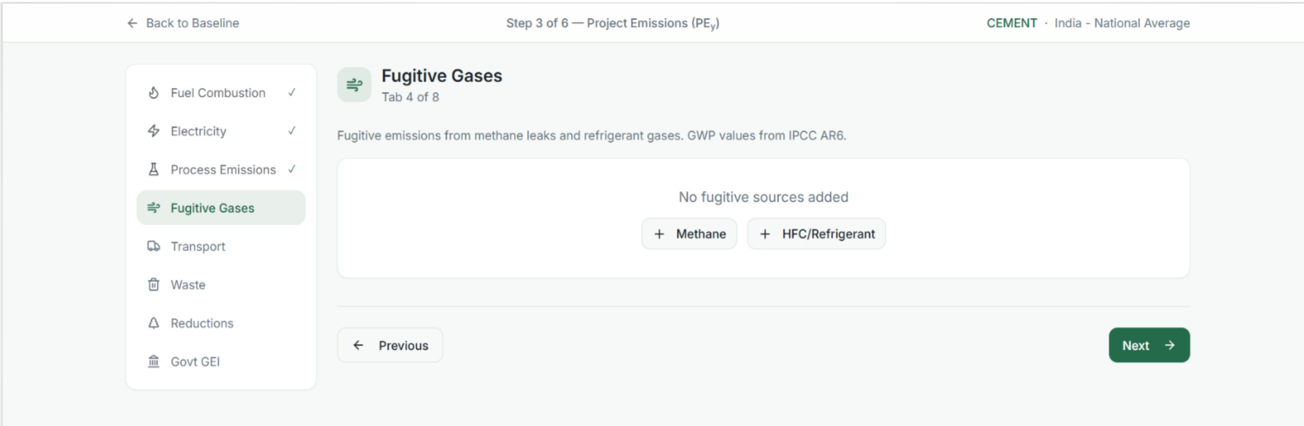
Field	What to Enter
Process (dropdown)	Select the process, e.g. 'Calcination (Preheater/Precalciner)'
Quantity (tonnes)	Tonnes of material processed (e.g., tonnes of limestone calcined)
EF (tCO_2 /t)	Pre-filled from IPCC. Edit only if you have certified data.
CO_2 (t)	Calculated automatically

Click '+ Add Process' to add more process types. Total Process CO_2 shows at the bottom.


← Back to Baseline
Step 3 of 6 — Project Emissions (PE_y)
CEMENT · India - National Average
Fuel Combustion ✓
Electricity ✓
Process Emissions
Fugitive Gases
Transport
Waste
Reductions
Govt GEI
Process Emissions
Tab 3 of 8
Process Flow & CO_2 Emission Zones
CEMENT MANUFACTURING — CO_2 EMISSION ZONES | Source: IPCC 2006, WBCSD/CSI GNR 2024
Hide
CO_2 Emission Zone
Normal Process Stage
Quarrying
Raw Mill
Preheater / Precalciner
Rotary Kiln ~1,450°C
Clinker Cooler
Finish Grinding
$CaCO_3 \rightarrow CaO + CO_2$
Kiln Fuel Combustion
EMISSION DETAILS
Preheater / Precalciner: Calcination of limestone — ~60% of total CO_2
Rotary Kiln ~1,450°C: Coal & Petcoke at ~1,450°C — ~40% of total CO_2
Red zones correspond to the emission sources in the input tabs below
Industry-specific process emissions for cement. Default EFs from IPCC are editable.
Process
Quantity (tonnes)
EF (tCO_2/t)
Calcination (Preheater/Precalci
7000
0.507
3,549 tCO_2
Kiln Fuel Combustion (Rotary Ki
7000
0.327
2,289 tCO_2
+ Add Process
Total Process CO_2
5,838 tCO_2
← Previous
Next →

Tab 4 — Fugitive Gases (Scope 1)

Fugitive emissions are accidental gas leaks methane from equipment, refrigerants from cooling systems, etc.



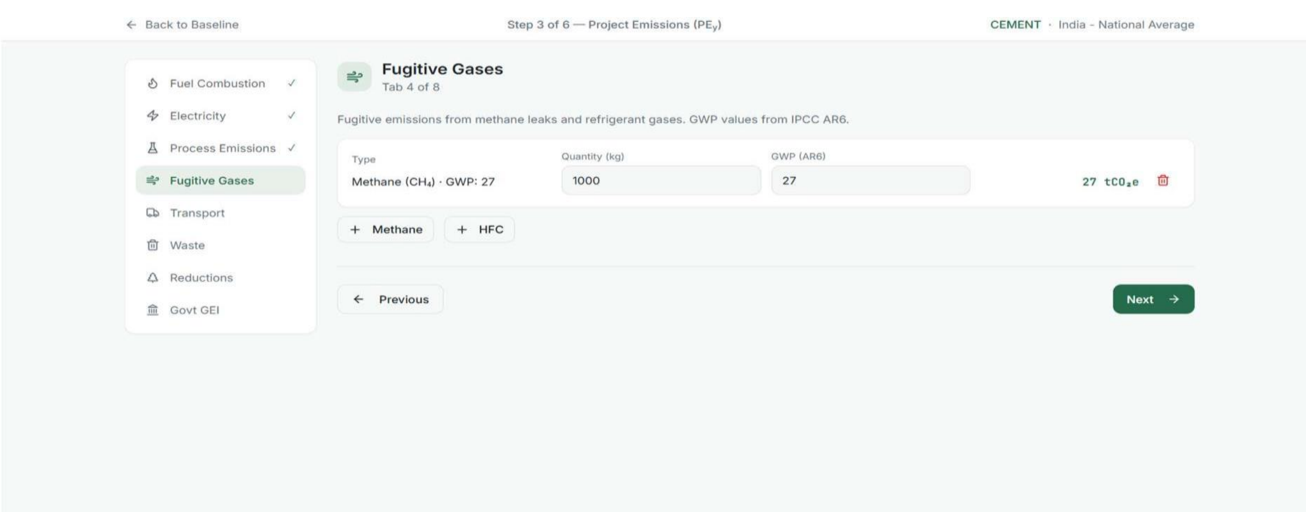
Also Called 'Purge Gas'

This tab may also include purge gas emissions gases intentionally released during start-up or maintenance of equipment. If your factory has these, add them here.

If you have no gas leaks or refrigerant use, click 'Next →' to skip this tab.

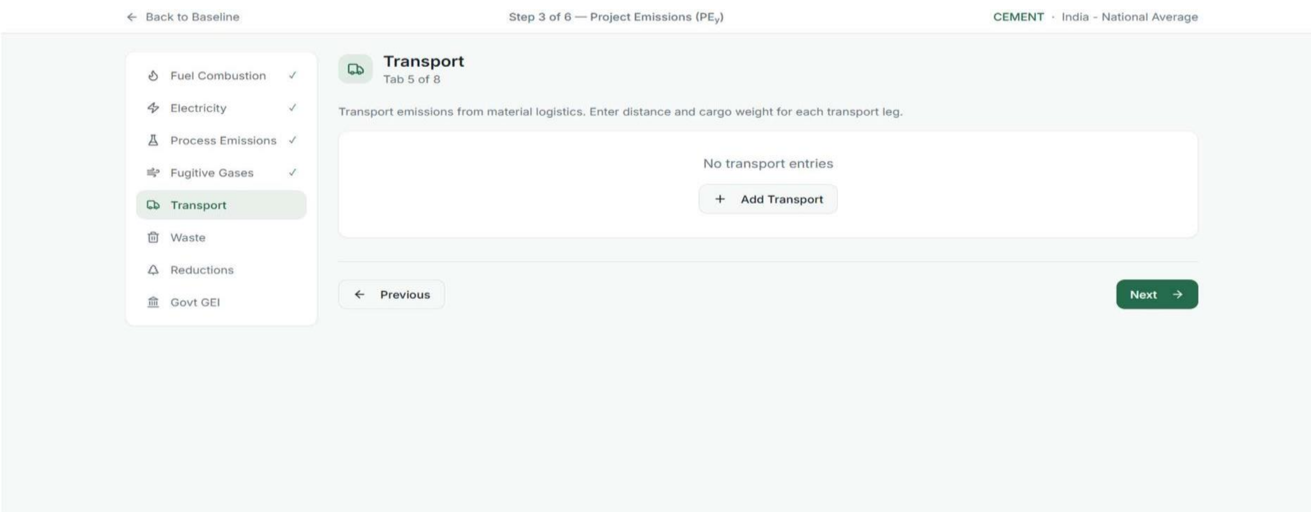
Otherwise, click '+ Methane' or '+ HFC/Refrigerant' and fill in:

Field	What to Enter
Type	Methane (CH_4) or HFC type (e.g., R-134a)
Quantity (kg)	Kilograms of gas leaked or released in the year
GWP (AR6)	Pre-filled — Global Warming Potential. Methane GWP = 27 (IPCC AR6)
tCO_2 e	Calculated automatically



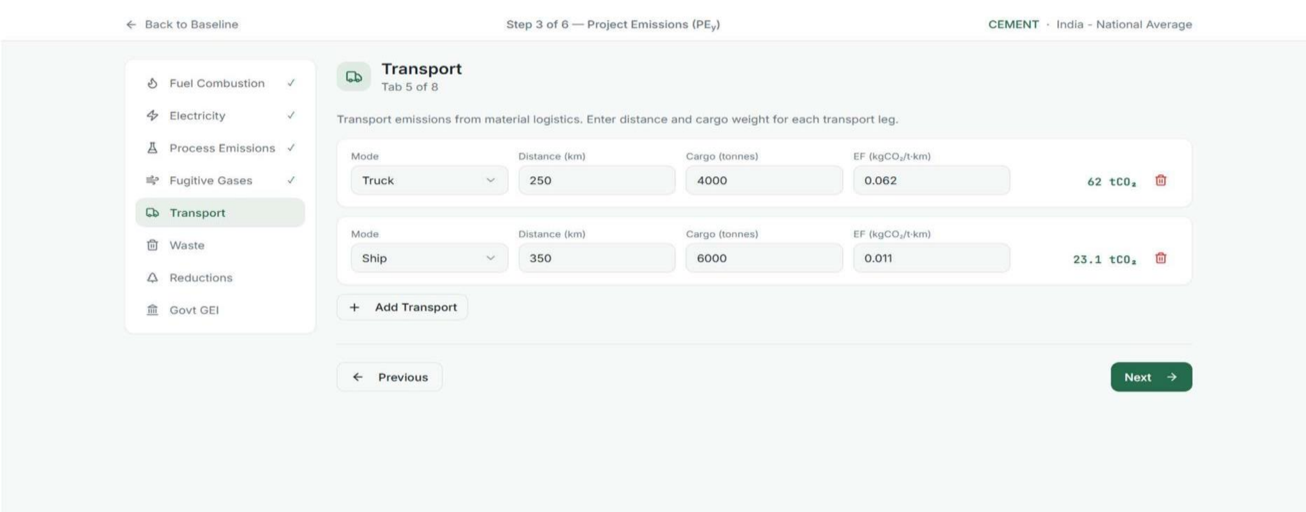
Tab 5 — Transport (Scope 3)

Emissions from transporting raw materials to your factory or finished products away from it.

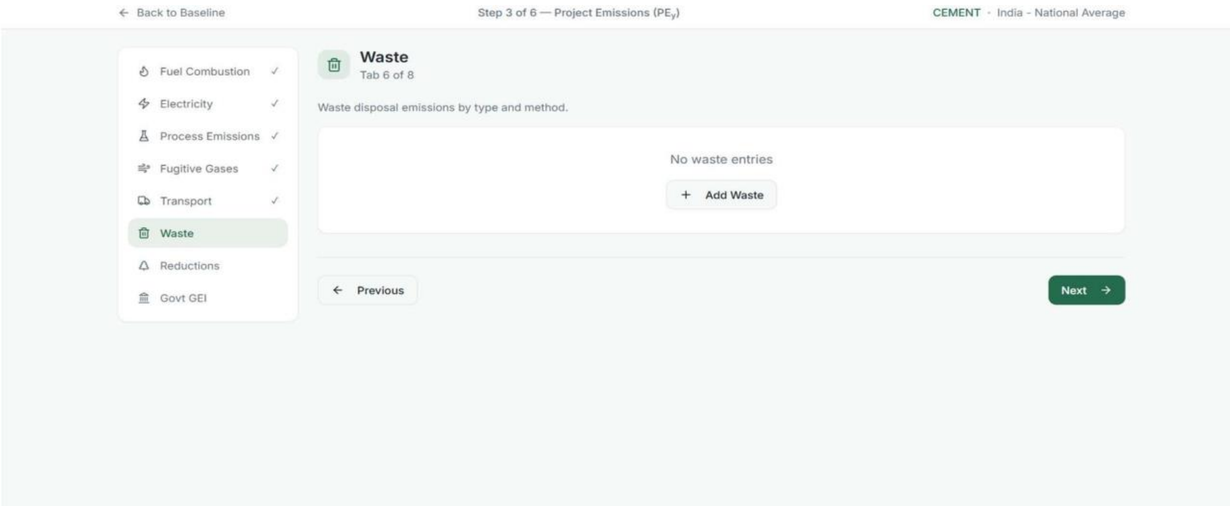


Click '+ Add Transport' for each transport leg, then fill in:

Field	What to Enter
Mode (dropdown)	Select: Truck, Ship, Rail, or Air
Distance (km)	Distance of the journey in kilometres
Cargo (tonnes)	Weight of goods transported in tonnes
EF ($kgCO_2$ /t-km)	Pre-filled based on transport mode. Truck = 0.062, Ship = 0.011
CO_2 (t)	Calculated automatically



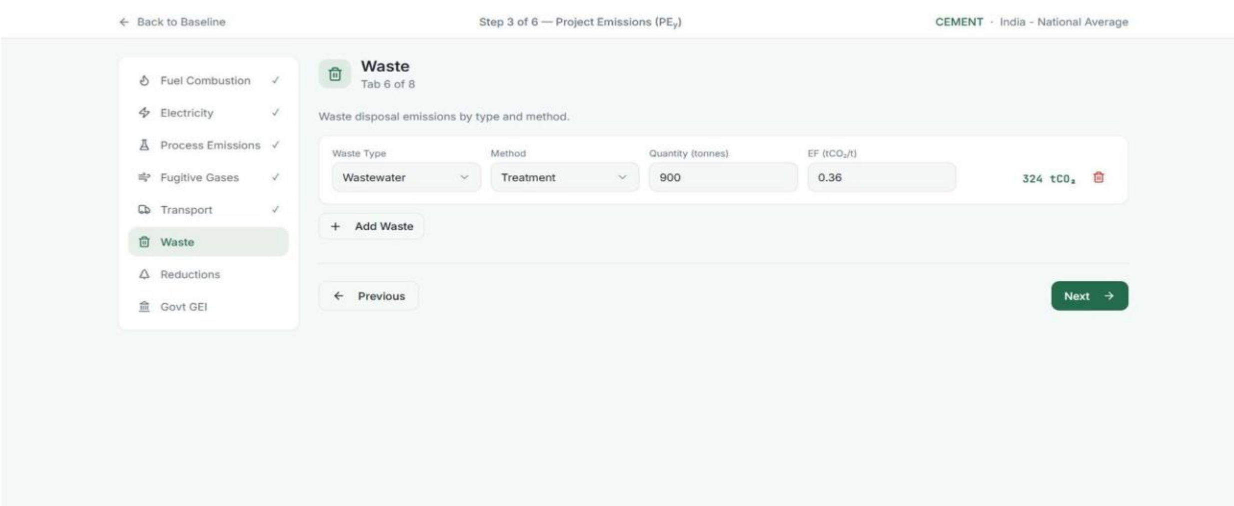
Tab 6 — Waste (Scope 3)



Emissions from how your factory disposes of waste materials.

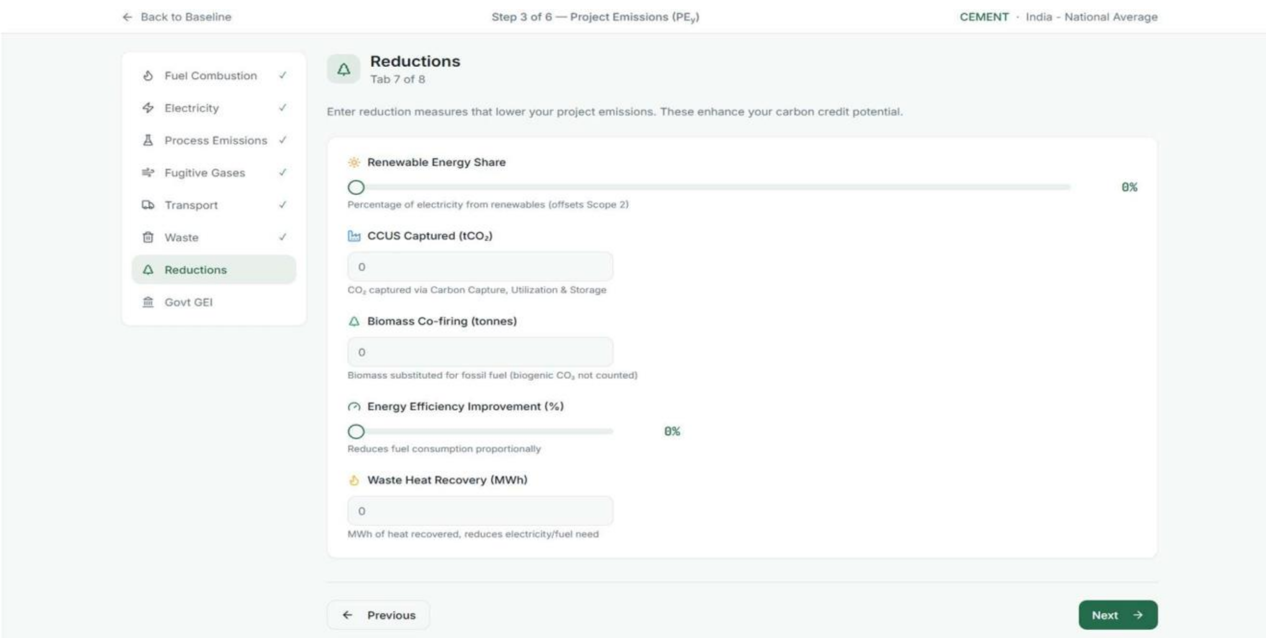
Click '+ Add Waste' for each waste stream:

Field	What to Enter
Waste Type (dropdown)	Wastewater, Solid Waste, Hazardous Waste, etc.
Method (dropdown)	How the waste is treated: Treatment, Landfill, Incineration, Recycling
Quantity (tonnes)	Tonnes of waste generated in the year
EF (tCO_2 /t)	Pre-filled from IPCC. Edit if you have certified data.

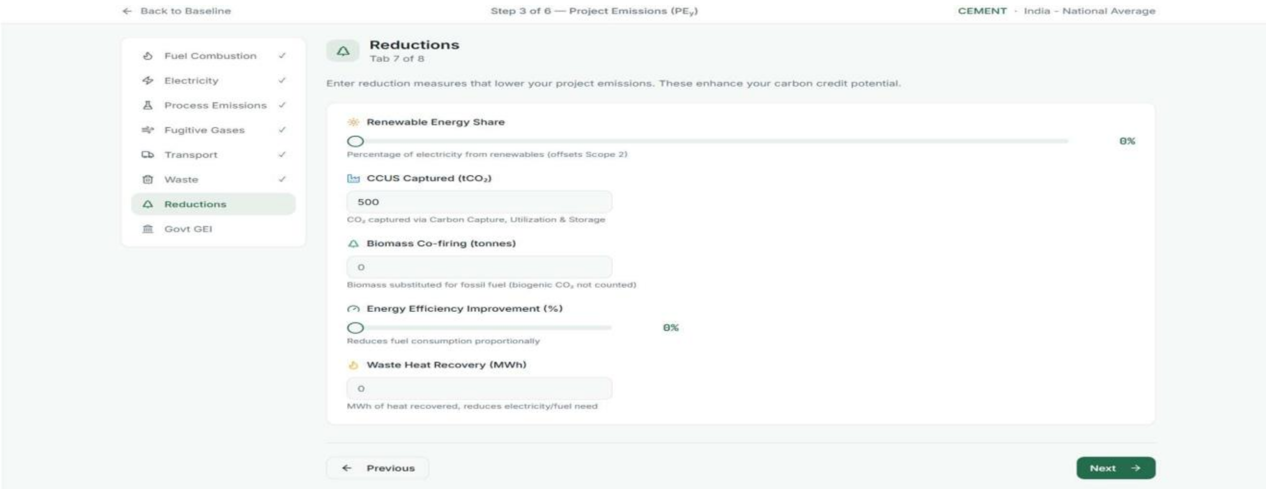


Tab 7 — Reductions

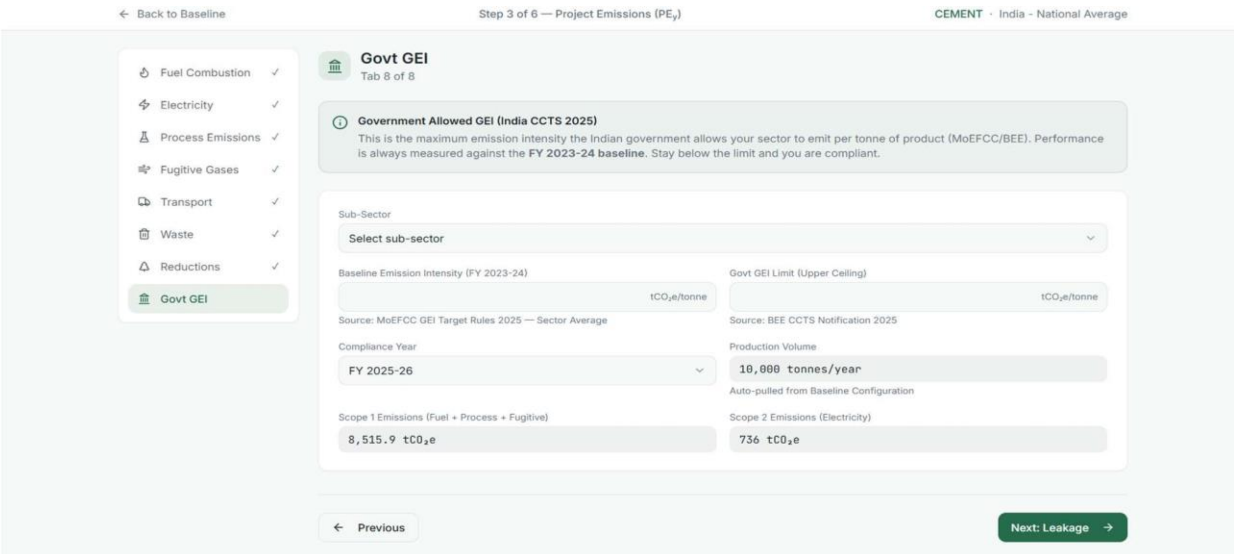
Enter all the green measures your factory has taken that REDUCE emissions. These improve your carbon credit score.



Field	What to Enter
Renewable Energy Share (%)	Percentage of electricity from solar, wind, or other renewables. Use the slider or type a number. Example: 20 means 20% of electricity is from renewables
CCUS Captured (tCO_2)	Tonnes of CO_2 captured by Carbon Capture, Utilisation & Storage systems, if any
Biomass Co-firing (tonnes)	Tonnes of biomass (agricultural waste, wood chips) used instead of coal
Energy Efficiency Improvement (%)	Percentage improvement in energy efficiency. This reduces fuel consumption proportionally.
Waste Heat Recovery (MWh)	Megawatt-hours of heat recovered from exhaust gases and reused


Back to Baseline
Step 3 of 6 — Project Emissions (PE_y)
CEMENT · India - National Average
Fuel Combustion
Electricity
Process Emissions
Fugitive Gases
Transport
Waste
Reductions
Govt GEI
Reductions
Tab 7 of 8
Enter reduction measures that lower your project emissions. These enhance your carbon credit potential.
Renewable Energy Share
0%
Percentage of electricity from renewables (offsets Scope 2)
CCUS Captured (tCO_2)
500
CO_2 captured via Carbon Capture, Utilization & Storage
Biomass Co-firing (tonnes)
0
Biomass substituted for fossil fuel (biogenic CO_2 not counted)
Energy Efficiency Improvement (%)
0%
Reduces fuel consumption proportionally
Waste Heat Recovery (MWh)
0
MWh of heat recovered, reduces electricity/fuel need
Previous
Next

Tab 8 — Govt GEI (Government Emission Intensity)



GEI stands for Greenhouse Gas Emission Intensity. This tab checks your factory's compliance with India's government targets under the Carbon Credit Trading Scheme (CCTS 2025).

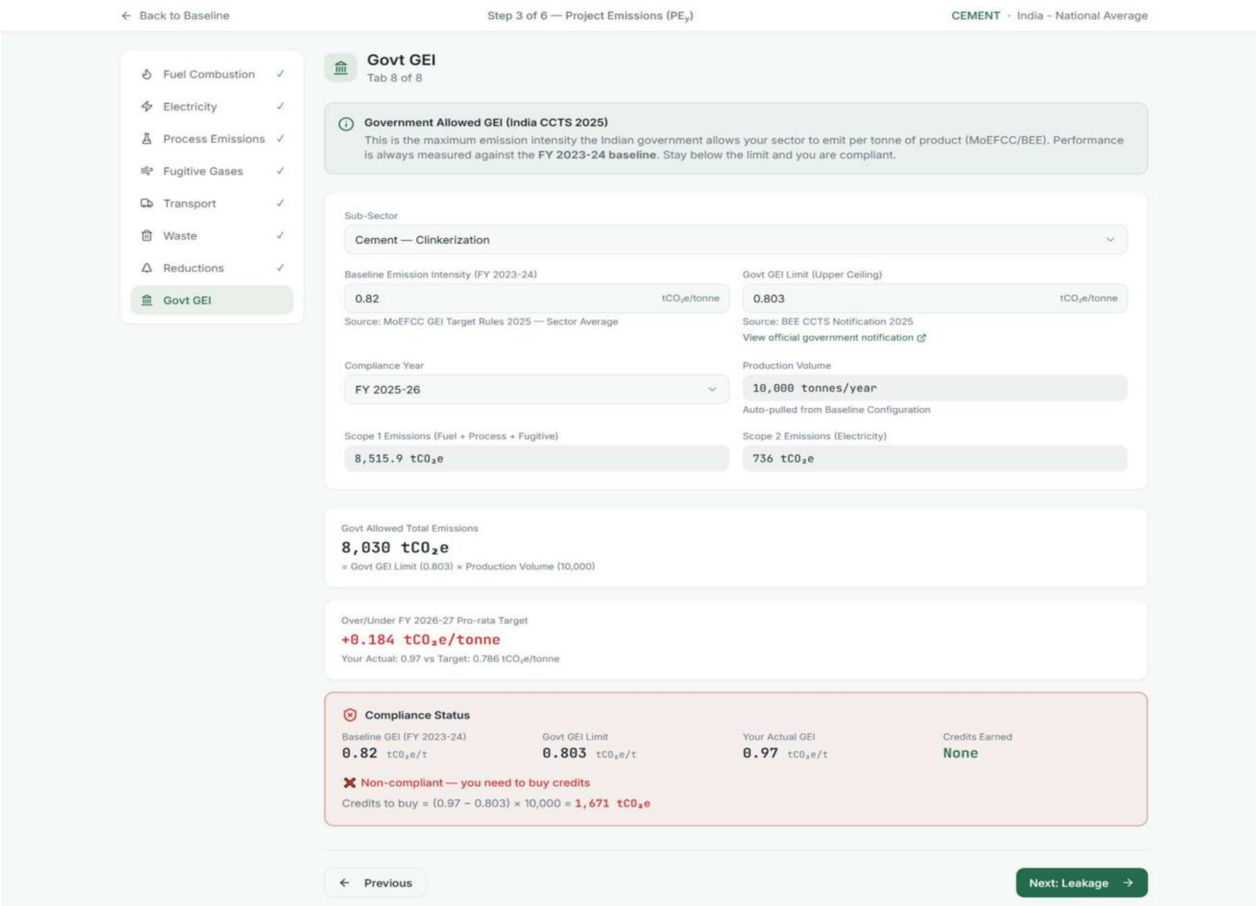
Field	What to Enter / Note
Sub-Sector (dropdown)	Select your specific sub-sector. Example for cement: 'Cement — Clinkerization'
Baseline Emission Intensity (FY2023-24)	Pre-filled from MoEFCC GEI Target Rules 2025. This is the government's reference number.
Govt GEI Limit (Upper Ceiling)	Pre-filled from BEE CCTS Notification 2025. This is the maximum you are allowed to emit per tonne.
Compliance Year	Select the year: FY2025-26
Production Volume	Auto-filled from your earlier entries — no need to change
Scope 1 & Scope 2 Emissions	Auto-filled from Tabs 1–6 — no need to change

Compliance Status (Appears After Filling Sub-Sector)

The calculator will show one of two results:

✅ COMPLIANT — Your emissions are below the government limit.
❌ NON-COMPLIANT — Your emissions exceed the limit. You need to buy credits.

Example (Non-Compliant): Actual GEI = 0.97 vs Govt Limit = 0.803 tCO_2 /tonne
Credits to buy = (0.97 − 0.803) × 10,000 = 1,671 tCO_2 e



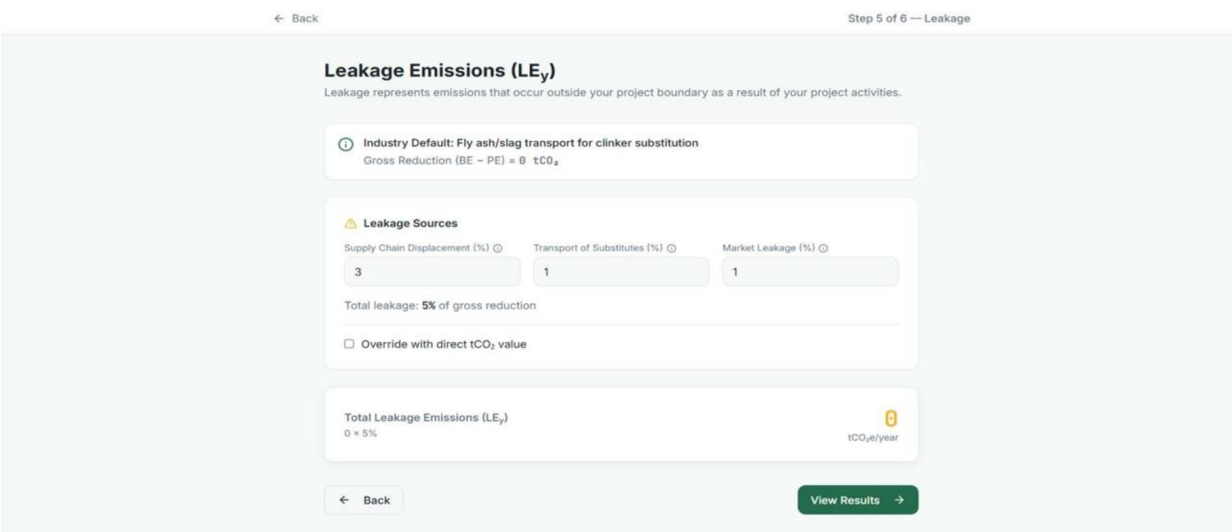
Click 'Next: Leakage →' to move to Step 5.

Step 5

Leakage Emissions

Account for emissions that happen OUTSIDE your factory boundary — Step 5 of 6

Leakage means CO_2 that is released somewhere else as a result of your project. For example, if your factory uses less clinker, someone else may produce more to compensate that is leakage.



What You See on the Leakage Page

Field	What It Means
Industry Default notice	Shows the leakage assumption pre-set for your industry. For cement, this is typically fly ash/slag transport.
Gross Reduction (BE − PE)	Calculated automatically — your total emission reduction so far
Supply Chain Displacement (%)	Percentage of your reduction that 'leaks' through supply chain changes. Pre-filled at 3%. Change only if you have specific data.
Transport of Substitutes (%)	CO_2 from transporting replacement materials. Pre-filled at 1%.
Market Leakage (%)	Market rebound effect. Pre-filled at 1%.
Total Leakage	Shows as '5% of gross reduction' — calculated automatically
Override with direct tCO_2 value	Tick this box ONLY if you have a precise leakage number from a certified audit

Most Users: Leave Leakage at Default Values

Unless you have a certified leakage measurement from an accredited auditor,
the pre-filled values (3% + 1% + 1% = 5%) are the industry standard.
Simply review the numbers and click 'View Results →'.

Step 6

Results Dashboard

View your carbon credit calculation — Step 6 of 6

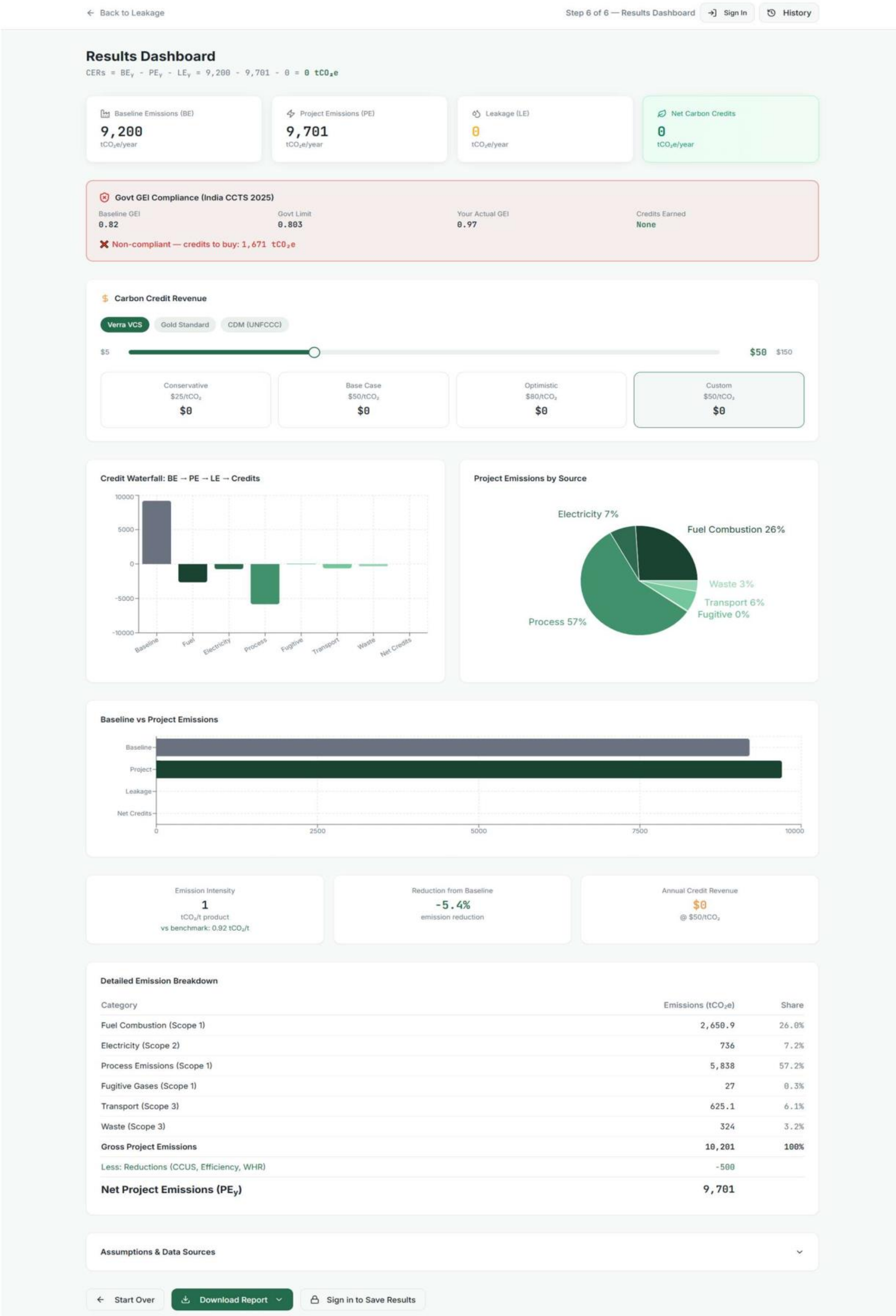
After clicking 'View Results', you will see the Results Dashboard. This is the final summary of all your calculations.

Understanding the Results Dashboard

Section	What It Shows
Formula bar at top: CERs = BEy − PEy − LEy	The complete calculation with your actual numbers filled in
Baseline Emissions (BE)	Your baseline number from Step 3 (e.g., 9,200 tCO_2 /year)
Project Emissions (PE)	Your actual emissions from Step 4 (e.g., 9,701 tCO_2 /year)
Leakage (LE)	Your leakage amount from Step 5
Net Carbon Credits (green box)	The final result — how many carbon credits you earned. Zero means you broke even.
Govt GEI Compliance box	Shows your compliance status (green = compliant, red = non-compliant)
Carbon Credit Revenue panel	Estimated money value of your credits at different carbon prices ($25, $50, $80/$tCO_2$)
Credit Waterfall Chart	A bar chart showing how each category contributed to your final credit number
Project Emissions by Source (pie chart)	Shows which category is your biggest source of emissions
Baseline vs Project bar chart	Visual comparison of before and after
Emission Intensity box	Your emissions per tonne of product vs the industry benchmark
Reduction from Baseline (%)	How much you reduced compared to baseline (negative = you emitted more)
Detailed Emission Breakdown table	Full table of every emission category with tonnes and percentage share
Assumptions & Data Sources	Click to expand and see all the standards and data sources used

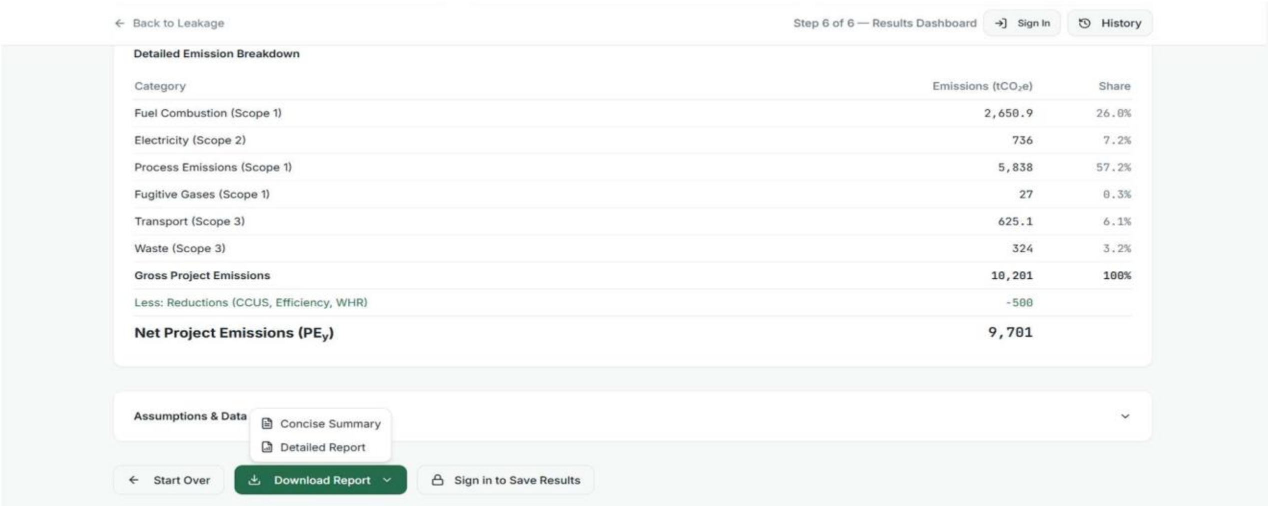
Example Results (Ultratech Cement, Clinkerization)

Baseline: 9,200 tCO_2 /year | Project Emissions: 9,701 tCO_2 /year
Leakage: 0 tCO_2 /year | Net Credits: 0 tCO_2 /year
Govt GEI Compliance: NON-COMPLIANT — Credits to buy: 1,671 tCO_2 e
Emission Intensity: 1.0 tCO_2 /product vs benchmark of 0.92 tCO_2 /t
Reduction from Baseline: −5.4% (meaning this factory emitted MORE than baseline)


Back to Leakage
Step 6 of 6 — Results Dashboard
Sign In
History
Results Dashboard
CERs = BEy - PEy - LEy = 9,200 - 9,701 - 0 = 0 tCO₂e
Baseline Emissions (BE)
9,200
tCO₂e/year
Project Emissions (PE)
9,701
tCO₂e/year
Leakage (LE)
0
tCO₂e/year
Net Carbon Credits
0
tCO₂e/year
Govt GEI Compliance (India CCTS 2025)
Baseline GEI
0.82
Govt Limit
0.803
Your Actual GEI
0.97
Credits Earned
None
Non-compliant — credits to buy: 1,671 tCO₂e
Carbon Credit Revenue
Verra VCS
Gold Standard
CDM (UNFCCC)
$5
$50
$150
Conservative
$25/tCO₂
$0
Base Case
$50/tCO₂
$0
Optimistic
$80/tCO₂
$0
Custom
$50/tCO₂
$0
Credit Waterfall: BE → PE → LE → Credits
10000
5000
0
-5000
-10000
Baseline
Fuel
Electricity
Process
Fugitive
Transport
Waste
Net Credits
Project Emissions by Source
Electricity 7%
Fuel Combustion 26%
Waste 3%
Transport 6%
Fugitive 0%
Process 57%
Baseline vs Project Emissions
Baseline
Project
Leakage
Net Credits
0
2500
5000
7500
10000
Emission Intensity
1
tCO₂/t product
vs benchmark: 0.92 tCO₂/t
Reduction from Baseline
-5.4%
emission reduction
Annual Credit Revenue
$0
@ $50/tCO₂
Detailed Emission Breakdown
Category
Emissions (tCO₂e)
Share
Fuel Combustion (Scope 1)
2,650.9
26.0%
Electricity (Scope 2)
736
7.2%
Process Emissions (Scope 1)
5,838
57.2%
Fugitive Gases (Scope 1)
27
0.3%
Transport (Scope 3)
625.1
6.1%
Waste (Scope 3)
324
3.2%
Gross Project Emissions
10,201
100%
Less: Reductions (CCUS, Efficiency, WHR)
-500
Net Project Emissions (PEy)
9,701
Assumptions & Data Sources
Start Over
Download Report
Sign in to Save Results

Downloading Your Report

At the bottom of the Results Dashboard, you will see a green 'Download Report' button with a small arrow (▾). Click the arrow to choose between two report types:



Option 1 — Concise Summary Report

A short, single-page PDF showing only the key numbers. Best for quick sharing with management or colleagues.

Contents of the Concise Report:

- Detailed Emission Breakdown table (all categories with tonnes and share %)
- Total Gross Project Emissions
- Net Project Emissions (PEy) after reductions

Annual Greenhouse Gas Emission Intensity Report

FY 2025-26 | Cement — Clinkerization

Entity Details

Company:	Ultratech Cement
Plant / Facility:	ADITYA CEMENT WORKS(Adityapuram Sawa Shambhupura Road Dist. Chittorgarh.
Reporting Year:	FY2025
Sub-Sector:	Cement — Clinkerization
Production Volume:	10,000 tonnes/year

GEI Summary Table

Metric	Baseline (FY 2023-24)	Current Achievement
Emission Intensity (tCO2e/t)	0.82	0.97
Govt GEI Limit (tCO2e/t)	—	0.803
Total Emissions (tCO2e)	8,200	9,701
Scope 1 (tCO2e)	—	8,516
Scope 2 (tCO2e)	—	736
Production Volume (t)	10,000	10,000

Over/Under FY 2026-27 Target: +0.184 tCO2e/t (Target: 0.786)

COMPLIANCE STATUS

NON-COMPLIANT - Entity exceeds Government GEI Limit

Carbon Credit Certificate (CCC) Eligibility

NOT ELIGIBLE - No over-performance against baseline.
Credits to purchase: 1,671 tCO2e

Digital Signature: ______________________________

Generated: 2026-04-15 07:04:11 UTC
Source: MoEFCC GEI Target Rules 2025 | BEE CCTS Notification 2025 | IPCC AR6
This is a system-generated report. Verify with BEE portal for official compliance status.

Option 2 — Detailed Report (Full PDF)

A comprehensive official document suitable for regulatory submissions, auditors, and board presentations.

Contents of the Detailed Report:

- Cover: Annual Greenhouse Gas Emission Intensity Report title
- Entity Details: Company, Plant, Reporting Year, Sub-Sector, Production Volume
- GEI Summary Table: comparing Baseline (FY2023-24) vs Current Achievement
- Compliance Status: shown in red/green with clear COMPLIANT or NON-COMPLIANT label
- Carbon Credit Certificate (CCC) Eligibility section
- Credits Earned or Credits to Purchase amount
- Over/Under FY2026-27 Pro-rata Target calculation
- Digital Signature line and generation timestamp
- Data Sources cited: MoEFCC GEI Target Rules 2025, BEE CCTS, IPCC AR6

Important Note on the Detailed Report

The detailed report states at the bottom:
'This is a system-generated report. Verify with BEE portal for official compliance status.'

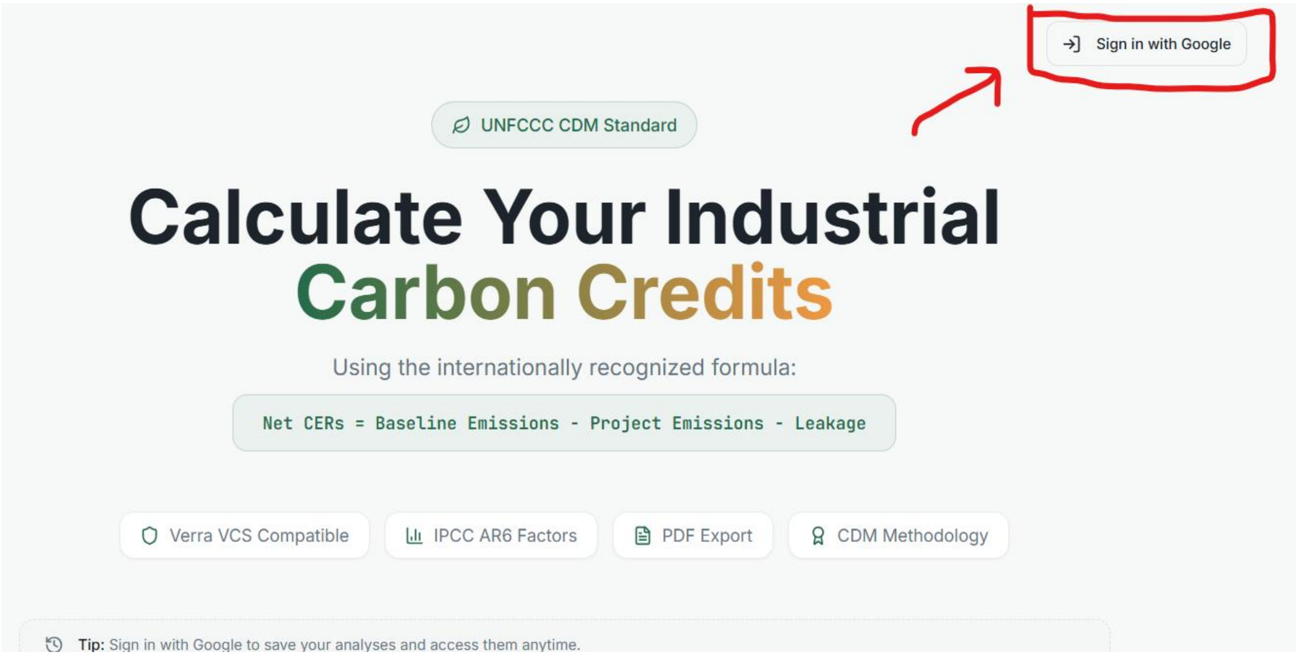
This means: this report is for reference and planning. For official government submission, you must cross-check with the BEE (Bureau of Energy Efficiency) portal.

The link of the detailed Report ids given below:

Detailed Report

Saving and Accessing Past Analyses

Your calculations can be saved for future reference but only if you sign in with Google.



How to Save Your Results

1. Click 'Sign in with Google' at the top-right of any page.
2. Choose your Google account and allow access.
3. Your current analysis is saved automatically once you are signed in.
4. You can also click 'Sign in to Save Results' at the bottom of the Results Dashboard.

Accessing Past Calculations

Click the 'History' button at the top-right of the Results Dashboard page. This shows all previously saved analyses linked to your Google account.

Why Sign In?

Without signing in: Your data lasts only while the browser tab is open.
With signing in: Your data is saved permanently and accessible from any device.
Signing in is free and uses your existing Google account.

Quick Reference — All 6 Steps at a Glance

Step	Section Name	Key Action
1	Industry Selection	Click your factory type, then click 'Start Calculator'
2	Company Setup	Enter company name, plant name, year, months, and select your region
3	Baseline	Choose baseline method and enter production volume
4	Project Emissions	Fill all 8 tabs: Fuel, Electricity, Process, Fugitive, Transport, Waste, Reductions, Govt GEI
5	Leakage	Review and accept the pre-filled leakage percentages (usually 5%)
6	Results	View dashboard, check compliance status, download report

Frequently Asked Questions

Q: I made a mistake in an earlier step. Can I go back?

Yes. Use the '← Back' button at the bottom of any page to return to the previous step. You can also click any tab on the left-side menu during Step 4 (Project Emissions) to jump directly to that tab.

Q: The emission factors (EF) are pre-filled. Should I change them?

Only change them if you have a certified lab test report or an auditor-provided value. The pre-filled values come from IPCC AR6, IEA, and India CEA — these are the internationally accepted standards. Changing them without certification may invalidate your report for official purposes.

Q: My factory uses multiple fuel types. How do I enter all of them?

In Tab 1 (Fuel Combustion), click '+ Add Fuel' once for each fuel type. Each fuel gets its own row. You can add as many fuels as needed — there is no limit.

Q: What does 'Non-Compliant' mean on the Govt GEI tab?

It means your factory's emissions per tonne of product (your Actual GEI) are higher than the government's allowed limit (Govt GEI Limit). This means your factory needs to purchase carbon credits to meet compliance. The exact number of credits to purchase is shown in the Compliance Status box.

Q: Can I use this report for official government submission?

The report is a calculation tool designed to help you prepare. The Detailed Report PDF states: 'This is a system-generated report. Verify with BEE portal for official compliance status.' For formal BEE/MoEFCC submissions, use this report as preparation and cross-check with the official BEE portal.

Q: What is the difference between Scope 1, Scope 2, and Scope 3?

Scope	What It Covers	Examples in This Calculator
Scope 1	Emissions from things your factory directly controls	Fuel combustion, process emissions, fugitive gas leaks
Scope 2	Emissions from electricity you buy from the grid	Electricity tab
Scope 3	Emissions from activities in your supply chain	Transport, Waste disposal

Glossary of Terms

Term	Simple Explanation
Baseline Emissions (BEy)	The CO_2 your factory 'should' have emitted without any green projects — the reference point
Project Emissions (PEy)	The CO_2 your factory actually emitted during the reporting year
Leakage (LEy)	CO_2 that is released somewhere else because of your factory's activities
Net CERs	Certified Emission Reductions — your final carbon credits. Positive = you earned credits.
GEI	Greenhouse Gas Emission Intensity — CO_2 emitted per tonne of product produced
Govt GEI Limit	The maximum GEI the Indian government allows for your sector (India CCTS 2025)
Grid EF	Grid Emission Factor — how much CO_2 was produced to generate 1 kWh of electricity
IPCC AR6	Sixth Assessment Report of the Intergovernmental Panel on Climate Change — the global scientific standard for emission factors
UNFCCC CDM	United Nations Framework on Climate Change — Clean Development Mechanism, the international framework for carbon credits
Verra VCS	Verified Carbon Standard — a leading private-sector carbon credit certification body
CCUS	Carbon Capture, Utilisation & Storage — technology that captures CO_2 before it reaches the atmosphere
tCO_2e	Tonnes of CO_2 equivalent — a standard unit for measuring greenhouse gases
Scope 1 / 2 / 3	Categories of emissions: direct (1), purchased energy (2), supply chain (3)
BEE	Bureau of Energy Efficiency — Indian government body managing the CCTS carbon trading scheme
MoEFCC	Ministry of Environment, Forest and Climate Change — Indian government ministry overseeing GEI rules

End of User Guide

Data sources: IPCC 2006 Guidelines · IEA 2016 · India CEA UCR CoU 2025 · GHG Protocol · IPCC AR6 GWP · World Steel Assoc. · GCCA